

Family Daayakattai (Dhayam) Mobile App

Complete Technical Specification, Senior-Friendly UX Engine, AI Coding Prompts & Deployment Blueprint

TARGET PLATFORMS	CORE AUDIENCE	COMMUNICATION	OPERATING COST
Android & iOS (Flutter)	Grandparents & Intergenerational	Agora RTC (Audio + 240p Video)	\$0 / month (0–10k Active Users)

1. Executive Product Overview & Problem Statement

Traditional South Indian **Daayakattai (Dhayam)** is an ancient 7×7 grid strategy game played across generations during family gatherings. While modern digital board game apps exist, they consistently fail elderly family members (grandparents) due to:

- **High Friction Onboarding:** Email logins, OTPs, passwords, and 6-character room PIN codes that older adults struggle to remember or type.
- **Aggressive Monetization & Dark Patterns:** Full-screen interstitial ads with tiny close buttons that trap seniors on unwanted web pages.
- **Disconnected Calling:** The separation of gameplay from video/voice calls (requiring multiple devices or clunky picture-in-picture modes).
- **Lack of Visual Guidance:** Unclear game states where older eyes cannot distinguish valid moves or understand entry requirements.

The Solution: A dedicated, ad-free, culturally authentic family communication app disguised as a traditional Daayakattai board. With one-tap WhatsApp deep links, call-style incoming push notifications, automatic video anchoring, and permanent one-time family profile setup, grandparents can play with their loved ones with zero technical stress.

2. Complete Zero-Cost Cloud Architecture

Layer	Recommended Technology	Tier & Quota	Role in Application
Cross-Platform Client	Flutter (Dart)	FREE & OPEN SOURCE	Single codebase rendering 60 FPS cloth board animations, audio FX, haptics, and WebRTC video tiles.
Realtime Database	Supabase / Firebase Spark	FREE (UP TO 50K MAU)	Low-latency WebSocket channel synchronizing dice rolls, board state, captured tokens, and active turns.
Voice / Video RTC	Agora RTC SDK	10,000 FREE MIN / MONTH	Provides crystal-clear audio and low-bandwidth 240p @ 15fps video streams anchored to player corners.
Call-Style Push Alerts	Firebase Cloud Messaging (FCM)	100% FREE UNLIMITED	Triggers full-screen incoming call ringtones on grandparents' phones ("Ananya is calling to play!").
Universal Deep Links	Cloudflare Pages	100% FREE	Hosts <code>assetlinks.json</code> and Apple Association files for one-tap WhatsApp link execution.

3. Daayakattai 7×7 Game Rules & State Machine

The game is played on a 7×7 grid (49 squares) using two 4-sided brass stick dice (*Kattai*) with dot values **0, 1, 2, 3**.

Core Game Mechanics

- **Entry Condition:** Coins start in home base. A roll of **1 (Dhayam)** is strictly required to move a coin out onto the player's home square.

Dice Roll Calculation & Bonus Rules

Dice Faces	Total Sum	Bonus Roll?
0 + 0	12 (Pannirendu)	YES
0 + 1	1 (Dhayam)	YES
0 + 2 / 1 + 1	2 (Rendu)	NO
0 + 3 / 1 + 2	3 (Moonu)	NO
1 + 3 / 2 + 2	4 (Naangu)	NO
2 + 3	5 (Ainthu)	YES
3 + 3	6 (Aaru)	YES

- **Outer Lap:** Coins traverse the outer perimeter (24 squares) counter-clockwise.
- **Malai (Safe Squares):** The 9 cross-marked squares (diamond pattern) are safe. Coins on safe squares cannot be cut.
- **Vettu (Capture):** Landing on an opponent on an unsafe cell captures their coin and sends it home.
- **Pazham (Inner Track & Goal):** A player must execute at least 1 **Vettu** before their coins can turn into the inner spiral lane toward the center goal (*Pazham*).

4. Master Prompts for AI Coding Assistants (Cursor / DeepSeek / Gemini)

Use these modular prompts sequentially in **Cursor IDE**, **DeepSeek R1**, or **Gemini CLI** to generate the complete codebase without writing code manually:

```
### PROMPT 1: PURE DART GAME ENGINE (daayakattai_engine.dart)
"You are a Senior Game Architect. Write a complete, standalone Dart engine for 7x7 Daayakattai without UI dependencies:
1. Dice Logic: Roll two 4-sided brass dice with sides [0, 1, 2, 3]. Sums: (0,0)->12, (0,1)->1, (0,2/1,1)->2, (0,3/1,2)->3, (1,3/2,2)->4, (2,3)->5, (3,3)->6. Rolls of 1, 5, 6, 12 grant an extra roll to a pending roll queue.
2. Track Coordinate Map: Model the 7x7 grid. Outer track has 24 cells. Safe cells (Malai) are at the 9 symmetric cross positions.
3. Movement Logic: Validate that un-entered pieces can only deploy on a roll of 1 (Dhayam). Track player cut history (hasVettu). Unlock inner Pazham track only if hasVettu == true.
4. Support 2-player and 4-player 2v2 partner team modes where team members cannot cut each other."
```

```
### PROMPT 2: SENIOR-CENTRIC FLUTTER BOARD UI (daayakattai_board.dart)
"Create a Flutter CustomPainter widget named `DaayakattaiBoardPainter` rendering a traditional South Indian 7x7 cloth board:
1. Visuals: Warm deep madder red (#581616) raw silk background with antique cream grid cells, brass-embossed borders, and hand-drawn lotus symbols on the 9 safe cross cells.
2. Piece Rendering: Render bell-shaped brass pawns with team-colored ring inlays (Red, Blue, Green, Yellow).
3. Senior Accessibility: When a roll is made, highlight all valid selectable pawns with a pulsating 3px emerald green glow. Tapping any piece automatically executes the legal move with smooth 300ms Tween animations.
4. Extra-Large Rolling Button: Render a 64dp high brass roll button across the bottom with Tamil and English labels."
```

```
### PROMPT 3: AGORA RTC VIDEO AVATAR INTEGRATION (agora_video_header.dart)
"Write a Flutter widget using `agora_rtc_engine` for 4-player family board games:
1. Layout: Place 4 circular video bubbles (radius 36dp) anchored near each player's board corner.
2. Low Bandwidth Mode: Configure video stream preset to 240p resolution at 15 FPS with audio-first priority to ensure seamless operation on budget smartphones and 3G/4G networks.
3. Active Speaker Feedback: Subscribe to `onAudioVolumeIndication` and display an animated glowing border around the speaking player. Include simple one-tap Mute Mic and Toggle Camera controls."
```

```
### PROMPT 4: WHATSAPP DEEP-LINKING & ZERO-CODE ONBOARDING (family_invite_service.dart)
"Write a Flutter service using `app_links`, `share_plus`, and `flutter_secure_storage`:
1. Generate Invite: Construct signed link: https://dhayam.app/join?room=ROOM_ID. Send via WhatsApp
intent: 'Namaste! Thatha is inviting you to play Daayakattai with video: https://dhayam.app/join?
room=ROOM_ID'.
2. Deep Link Handler: On launch, parse `room` query parameter and navigate straight to active GameView,
bypassing all login screens.
3. Delegated Profile: Save display name and pinned family circle in local secure storage permanently so
older users are never logged out."
```

5. Nano Banana Asset Generation Prompts

Pass these prompts into **Google AI Studio / Gemini 3 Image (Nano Banana Pro)** to generate all authentic graphic assets:

Asset 1: 7x7 Traditional Embroidered Board (Top-Down Orthographic)

"Top-down orthographic 2D view of a traditional South Indian Daayakattai 7x7 square cloth board game (Kattam). Woven heavy raw silk texture in deep rustic madder red (#581616) and antique cream (#FFF8E7). Intricate hand-embroidered golden borders. Clear 7x7 grid lines with cross-hatch 'X' motifs embroidered on the 9 sacred safe squares (Malai/Kottai) and a radiating lotus mandala embroidered in the center square (Pazham). High contrast, authentic folk art detailing, flat lay, sharp focus, 4K resolution, 1:1 aspect ratio."

Asset 2: 4-Sided Brass Stick Dice (Kattai Pair)

"Product photography of a pair of traditional South Indian Daayakattai solid brass stick dice (Dhayam Kattai). Long rectangular cuboid form with beveled corners, warm polished antique brass patina with dark engravings. Face 1 shows 1 stamped circular dot; Face 2 shows 2 dots; Face 3 shows 3 dots; Face 4 is blank. Resting on dark teak wood surface, casting soft shadows. Photorealistic, macro studio lighting, 8k resolution, 16:9 aspect ratio."

Asset 3: Conical Brass Pawns (Kaai)

"Game asset set of 4 traditional South Indian brass board game pawns (Daayakattai Kaai). Distinctive miniature temple gopuram and bell-shaped lathe-turned solid brass tokens with colored lacquer inlay rings: Ruby Red, Emerald Green, Royal Cobalt Blue, and Deep Saffron Yellow. Clean matte finish, isometric view, uniform soft shadow, isolated on neutral warm beige background, game-ready UI sprite sheet, 4K resolution."

6. Complete Database Schema (Supabase PostgreSQL)

```
-- SUPABASE DATABASE SETUP SCRIPT
CREATE TABLE public.game_rooms (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  created_at TIMESTAMP WITH TIME ZONE DEFAULT timezone('utc'::text, now()) NOT NULL,
  status TEXT DEFAULT 'waiting' CHECK (status IN ('waiting', 'in_progress', 'completed')),
  active_player_index INT DEFAULT 0,
  pending_roll_queue INT[] DEFAULT '{}',
  board_state JSONB NOT NULL DEFAULT '{
    "players": [
      {"id": "p1", "name": "Thatha", "team": 1, "tokens": [-1,-1,-1,-1], "hasVettu": false},
      {"id": "p2", "name": "Rohan", "team": 2, "tokens": [-1,-1,-1,-1], "hasVettu": false},
      {"id": "p3", "name": "Paati", "team": 1, "tokens": [-1,-1,-1,-1], "hasVettu": false},
      {"id": "p4", "name": "Maya", "team": 2, "tokens": [-1,-1,-1,-1], "hasVettu": false}
    ],
    "safe_squares": [3, 9, 15, 21, 27, 33, 39, 45, 24]
  }',
  agora_channel_token TEXT
);

-- Enable Realtime WebSockets on game_rooms
ALTER PUBLICATION supabase_realtime ADD TABLE public.game_rooms;
```

7. Testing & Quality Assurance Plan

1. Low-End Hardware & Thermal Testing

Test on entry-level Android devices (2GB–3GB RAM, budget MediaTek/Snapdragon chipsets). Verify that Flutter renders at a stable 60 FPS and that Agora 240p video calling does not cause thermal throttling or rapid battery drain.

2. Network Degradation & Audio Resilience

Simulate throttled 3G networks and 20% packet loss. Ensure the app prioritizes voice audio stream uninterrupted, automatically dropping video frames if necessary without disconnecting the match.

3. Grandparent "10-Second" Usability Benchmark

Conduct live testing with non-tech-savvy seniors over 60 years old. A grandparent should be able to accept an incoming invite and roll the dice within 10 seconds of receiving a notification, without asking for verbal assistance.

8. Publishing Requirements & 1st-Year Budget Summary

Publishing Steps

- **Google Play Store:** \$25 one-time developer registration. Requires completing a 14-day closed test with 12–20 testers for new personal accounts.
- **Apple App Store:** \$99/year Apple Developer Program. Requires privacy justification strings for Camera and Microphone usage in Info.plist.
- **Privacy Policy:** Free hosting on Cloudflare Pages explaining that camera/mic streams are peer-to-peer ephemeral and never recorded.

Cost Summary Matrix

Item	Upfront	Monthly
Flutter & VS Code / Cursor	\$0	\$0
Supabase Realtime (Free Tier)	\$0	\$0
Agora Video (10k min/mo)	\$0	\$0
Domain & Cloudflare Links	~\$10	\$0
Google Play License	\$25	\$0
Apple Developer License	\$99	\$0
TOTAL FIRST YEAR	~\$134	\$0.00

9. First-Weekend Quickstart Execution Checklist

1. **Setup Flutter & Cursor IDE:** Run `flutter create --org com.dhayam daayakattai_app`.
2. **Generate Engine:** Paste **Prompt 1** into DeepSeek R1 to generate `daayakattai_engine.dart`.
3. **Generate UI:** Paste **Prompt 2** into Claude 3.7 / Cursor to generate the `DaayakattaiBoardPainter`.
4. **Generate Graphic Assets:** Copy the 3 **Nano Banana** prompts into Google AI Studio to get high-res board textures, brass dice, and pawns.
5. **Wire Agora & Supabase:** Paste **Prompts 3 & 4** to enable video calling and WhatsApp invites. Run `flutter run` on your phone!